

## Gang Jiang

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### RESEARCH INTERESTS

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AI-Driven Automated Modeling; Large Language Models; High-Performance Computing;  
Building Performance Simulation; Sustainability & Resilience; Smart & Healthy Built Environment

### EDUCATION

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- **University of Utah** Salt Lake City, UT, USA  
*Ph.D. in Civil Engineering* Aug. 2022 – Jun. 2026 (*Expected*)
- **Tianjin University** Tianjin, China  
*M.Sc. in Architecture & Building Sciences, **Excellent Thesis*** Aug. 2019 – Jun. 2022
- **Yancheng Institute of Technology** Yancheng, Jiangsu, China  
*B.Sc. in Heating, Ventilation, & Air Conditioning* Aug. 2015 – Jun. 2019

### RESEARCH EXPERIENCE

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- **University of Utah, Department of Computer Science** Salt Lake City, UT  
*Research Assistant* May. 2025 - Present
  - **US-NSF Projects:** (Primary Contributor) **Large language model (LLM) testing and development:** Conducted large-scale evaluation of LLMs, designed benchmarking protocols, and contributed to refining LLM architectures and training strategies for improved accuracy, efficiency, and robustness in specific domains.
- **University of Utah, Department of Civil & Environmental Engineering** Salt Lake City, UT  
*Research Assistant* Aug. 2022 - May. 2025
  - **US-NSF #2311685:** (Primary Contributor) Developed LLM-driven **auto-building energy modeling platforms (EPlus-LLM v1/v2, and Prompt-ABEM)** by integrating natural language with physics-based simulation to **enhance the accessibility and scalability of modeling.**
  - **US-NSF #2318720:** (Proposal Drafting, Primary Contributor) Drafted sections on the co-simulation framework and experimental design; Conducted **co-simulations of thermo-responsive desiccants** for building dehumidification and decarbonization, using EnergyPlus, Modelica, and Ansys, to **support healthy and sustainable built environments.**
  - **Global Change & Sustainability Center:** (Primary Contributor) Developed **high-fidelity building energy models** (by Resistance-Capacity mechanism and EnergyPlus software) and **efficient auto-calibration framework** that integrates an enhanced Bayesian inference method with a deep-learning, high-resolution surrogate model, facilitating the development of digital twins for **real-time monitoring and fault detection** in real-world.
  - **US-Utah-DOT #24-8342:** (Primary Contributor) Designed a **scalable Vision-AI tool** for automated road snow condition assessment, enabling more **effective snowplow operations and enhanced traffic safety.**
  - **US-Utah-DOT #24-8332:** (Contributor) Volumetric measurement of salt stockpiles using **photogrammetry, LiDAR, and depth cameras:** Reconstructed 3D point cloud models of all stockpiles in CloudCompare and conducted volume calibration and results evaluation.
- **Tianjin University** Tianjin, China  
*Research Assistant* Aug. 2019 - Jun. 2022
  - **China-NSFs:** Developed **fault detection & diagnosis methods for smart building systems** with incomplete data, integrating Modelica simulations with machine learning to enhance accuracy and robustness.

- **Industry Projects:** Conducted research on building energy efficiency and management, focusing on **optimizing energy efficiency and sustainability strategies**.

## • Amazon Web Services (AWS)

Beijing, China

Intern

Jun. 2021 - Dec. 2021

- **Data Center Design:** Assisted in **planning and optimizing** data center infrastructure, including local generators, uninterruptible power supplies (UPS), power distribution, and cooling systems to enhance **resilience and scalability**.
- **Data Center Operation:** Contributed to improving **energy efficiency and fault detection** in data centers through operational optimizations and system monitoring .

## JOURNAL PUBLICATIONS

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- [8] [G. Jiang](#), K. Chen, et al. Benchmarking Knowledge and Capability of Large Language Models in Building Science Domain. *Energy Use, The Innovation*, Dec. 2025.
- [7] [G. Jiang](#), J. Chen. Efficient Fine-Tuning of Large Language Models for Automated Building Energy Modeling in Complex Cases. *Automation in Construction*, Jul. 2025.
- [6] Z. Ma, [G. Jiang](#), J. Chen. A Neural Ordinary Differential Equations-Based Approach for Enhanced Building Energy Modeling on Small Datasets. *Building Simulation*, Apr. 2025.
- [5] Z. Ma, [G. Jiang](#), Y. Hu, J. Chen. A Review of Physics-Informed Machine Learning for Building Energy Modeling. *Applied Energy*, Mar. 2025.
- [4] [G. Jiang](#), Z. Ma, L. Zhang, J. Chen. Prompt Engineering to Inform Large Language Models in Automated Building Energy Modeling. *Energy*, Feb. 2025. 🏆 ESI - Top 1% Highly Cited. [G Google Scholar](#).
- [3] Z. Ma, [G. Jiang](#), J. Chen. Physics-Informed Ensemble Learning with Residual Modeling for Enhanced Building Energy Prediction. *Energy and Buildings*, Nov. 2024.
- [2] [G. Jiang](#), Y. Chen, Z. Wang, K. Powell, B. Billings, J. Chen. A Deep Learning-Based Bayesian Framework for High-Resolution Calibration of Building Energy Models. *Energy and Buildings*, Nov. 2024.
- [1] [G. Jiang](#), Z. Ma, L. Zhang, J. Chen. EPlus-LLM: A Large Language Model-Based Computing Platform for Automated Building Energy Modeling. *Applied Energy*, Aug. 2024. 🏆 ESI - Top 1% Highly Cited. [G Google Scholar Cited 100+ times](#).

## PROJECT REPORTS

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- [3] M. Abbas, M. Pouria, K. Biao, [G. Jiang](#), R. Abbas, J. Chen, W. Jessica. Volumetric Measurement of Salt Piles Using Photogrammetry, LiDAR, and Depth Cameras. *Utah Department of Transportation*, Apr. 2025.
- [2] [G. Jiang](#), B. Kuang, Z. Shi, Z. Ma, J. Chen. Snow Plow Performance Measures in Non-RWIS Locations. *Utah Department of Transportation*, Aug. 2024.
- [1] [G. Jiang](#), J. Chen, P. Kody, B. Blake. High-Fidelity Building Modeling Enabled Digital Twin Development for Continuous Building and HVAC Monitoring. *SEED2SOIL Programe, Global Change & Sustainability Center*, Aug. 2023.

## CONFERENCE PUBLICATIONS

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- [3] G. Jiang, J. Chen. Automated Building Energy Modeling Using Large Language Model. *COBEE*, Jul. 2025.
- [2] G. Jiang, J. Chen. Fine-Tuning Large Language Models for Automated Building Simulation. *BAS*, Jul. 2025.
- [1] [G. Jiang](#), L. Zhang, J. Chen. EPlus-LLM: A Novel Automated Building Simulation Platform Using Natural Language Processing. *ASHRAE Annual Conference*, Jul. 2024.

## TEACHING & MENTORING EXPERIENCE

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- **Teaching Assistant:** [Green Building Course] Prepared class materials, slides, and the syllabus; held weekly office hours; assisted students with project assignments.
- **Computer Lab Instructor:** [Green Building Course] Led a 3-week hands-on module on building energy modeling using EnergyPlus & OpenStudio, focusing on sustainable building design, retrofit analysis, and energy efficiency.
- **Research Mentorship:** [Our Research Group] Mentored undergraduate and early-stage PhD students through weekly meetings on research framing, data processing, and AI workflow development.

## CONFERENCE & WORKSHOP PRESENTATIONS

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- **ASHRAE Winter:** Transforming Building Energy Modeling Processes with Large Language Models. Feb. 2026. [Invited Talk](#)
- **ASHRAE Winter:** Real-World Applications of the EPlus-LLM platform. Feb. 2026. [Poster](#)
- **Build Next:** Toward Automated Building Energy Modeling with Large Language Models. Oct. 2025. [Invited Talk](#)
- **ASHRAE CIDCO:** Automating Building Energy Modeling from Natural Language. Aug. 2025. [Invited Talk](#)
- **ASHRAE Annual:** EPlus-LLM: A Novel Automated Building Simulation Platform Using Natural Language. Jun. 2024. [Oral Presentation](#)
- **GCSC Environment and Sustainability Research Symposium:** Improved Bayesian Calibration of Campus Building Energy Models. Feb. 2023. [Poster](#)

## OPEN-SOURCE CONTRIBUTIONS

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- **EPlus-LLM:** LLM-driven automated building energy modeling through natural language. ([HuggingFace](#); [GitHub](#))
- **EPlus-LLMv2:** New version of EPlus-LLM for more complex building modeling scenarios, e.g., building geometries, operation settings, and human behaviors. ([HuggingFace](#); [GitHub](#))
- **Prompting LLMs for ABEM:** A comprehensive guideline for prompt engineering of LLMs in automated building energy modeling for real-world applications. ([GitHub](#))

## SERVICES

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- **GitHub Maintainer:** I maintain open-source models/repositories (have been downloaded hundreds of times), and actively respond to issues.
- **Independent Reviewer:** *Renewable & Sustainable Energy Reviews; Cell Reports Physical Science; Building Simulation*
- **ASHRAE:** Associate Member

## PROGRAMMING SKILLS

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- **Programming Languages:** Python, PyTorch, JavaScript
- **Technologies:** High-Performance Computing on Linux, Model Deployment, GUI Development
- **Software:** EnergyPlus, SketchUp, Matlab, Dymola (Modelica)